

EMCH 260

SOLID MECHANICS

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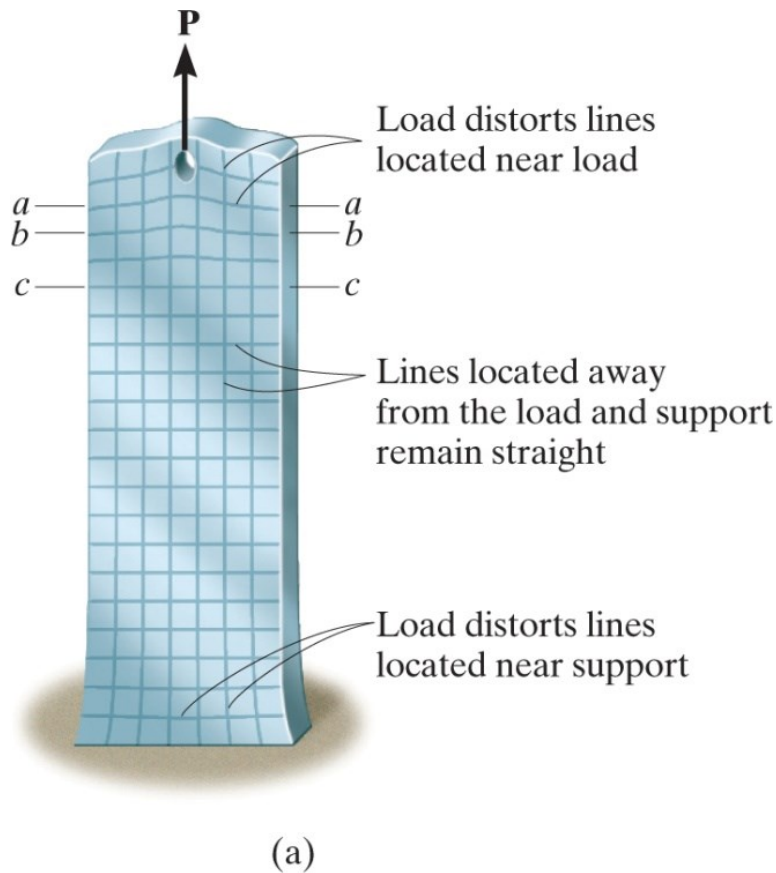
Axial Load

4

Chapter 4.1

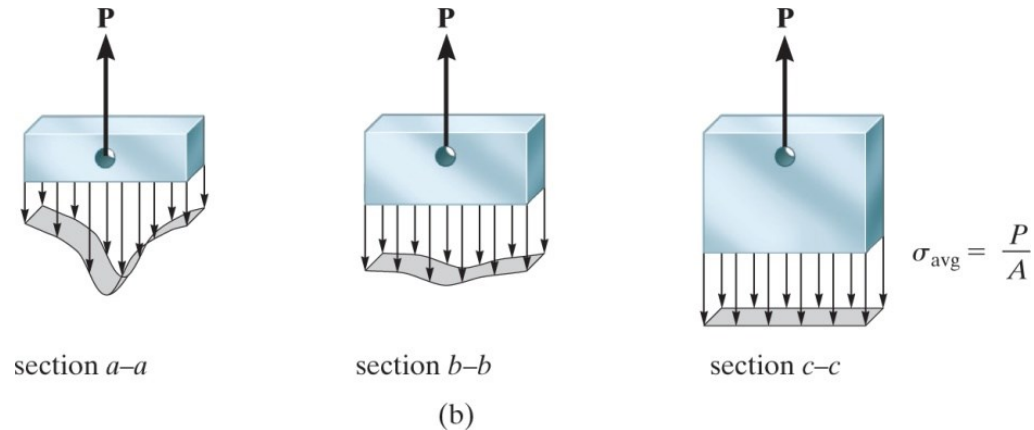
- ✓ Understand the Saint-Venant's principle
- ✓ Determine the elastic deformation of axially loaded member

LOCALIZED DEFORMATION



For any section perpendicular to external force direction

$$P = \int dP = \int_A \sigma dA \quad \sigma_{\text{avg}} = \frac{P}{A}$$

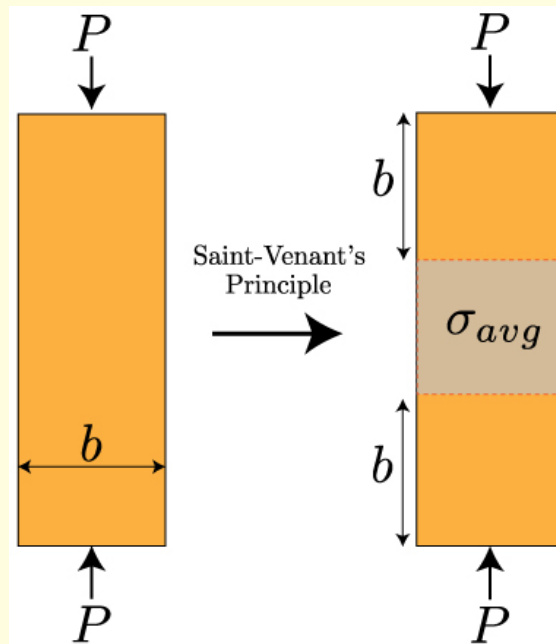


For different section the spatial distribution of stress in the section plane could be different.

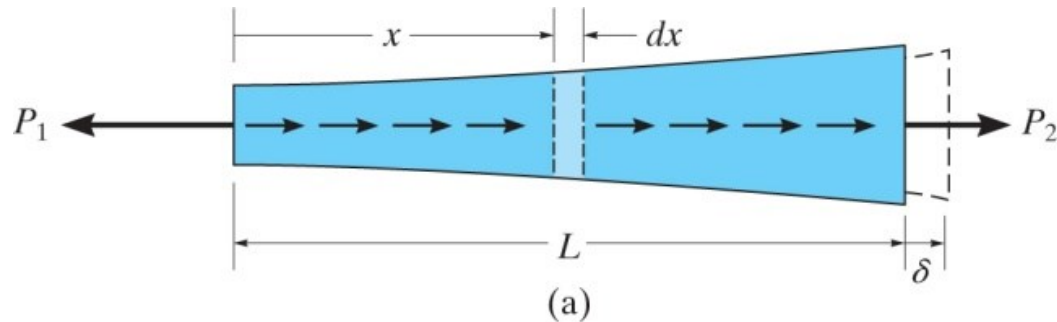
- **Localized deformation** occurs at the ends where external loadings or support and corresponding spatial stress distribution is not uniform (section *a-a* and *b-b*)
- At a distance at least equal to the largest dimension of the loaded cross section, the deformation and stress become spatially uniform (section *c-c*)

SAINT-VENANT'S PRINCIPLE

The stress distributions at different cross sections are different. However, at locations far enough away from the support and the applied load, the stress distribution becomes uniform. The fact that the localized stress and deformation behave in this manner is called **Saint-Venant's Principle** (Saint-Venant, 1855).

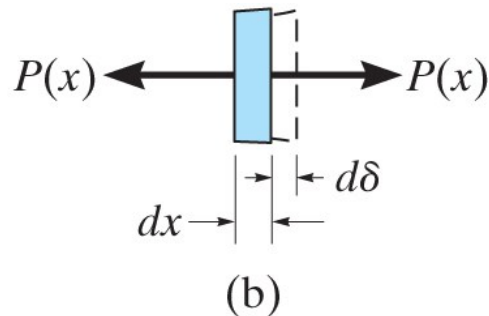


ELASTIC DEFORMATION OF AN AXIALLY LOADED MEMBER



Concentrated loads (P_1 and P_2) and variable external load distributed along the length

Free-body diagram of any small element perpendicular to the axial direction



$$\sigma = \frac{P(x)}{A(x)} \quad \text{and} \quad \varepsilon = \frac{d\delta}{dx}$$

Provided these quantities do not exceed the proportional limit, we can relate them using **Hooke's Law**, i.e. $\sigma = E \varepsilon$, therefore

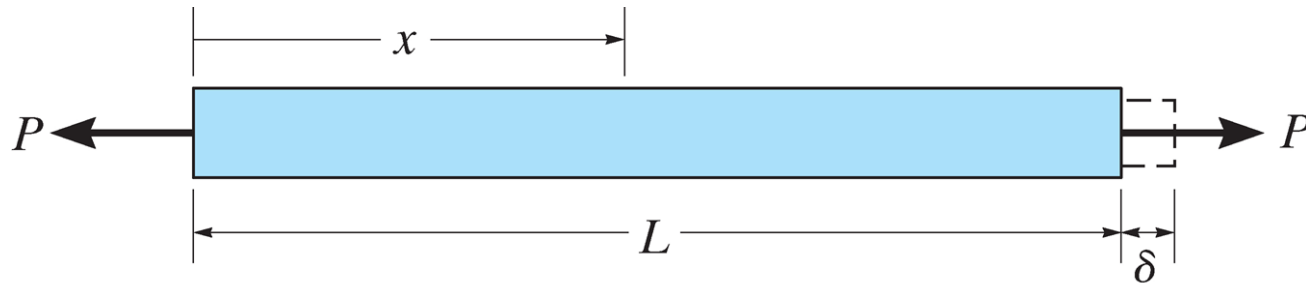
$$\frac{P(x)}{A(x)} = E(x) \left(\frac{d\delta}{dx} \right)$$

$$d\delta = \frac{P(x)dx}{A(x)E(x)}$$

$$\delta = \int_0^L \frac{P(x)dx}{A(x)E(x)}$$

CONSTANT LOAD & CROSS-SECTIONAL AREA

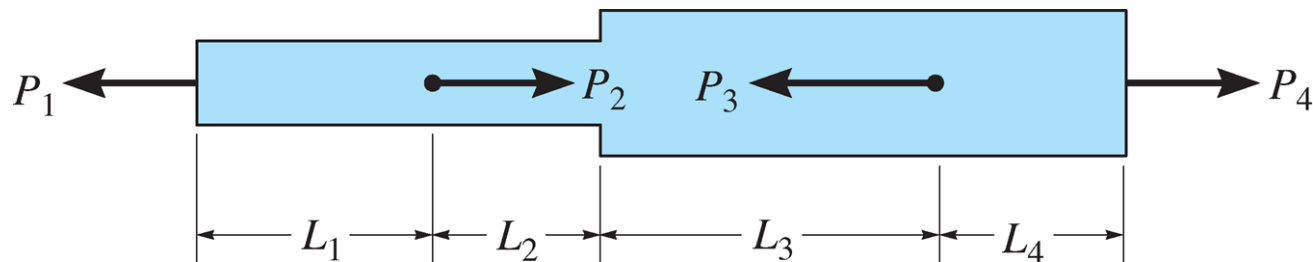
For Constant cross-sectional area A and the homogenous material (i.e. constant E), these two factors can be taken out of the integral. Furthermore, if a constant force is applied at each end, then the internal force N throughout the material is also constant.



$$\delta = \int_0^L \frac{P(x)dx}{A(x)E(x)} = \frac{1}{AE} \int_0^L P(x)dx = \frac{PL}{AE}$$

DIFFERENT AXIAL LOADS & ABRUPTLY CHANGED CROSS-SECTIONAL AREA

For several different axial forces along the axial direction, and/or the cross-sectional area and/or modulus of elasticity changes abruptly from one region to another, we can consider each segment as constant load and constant cross-sectional area. The total displacement of one end with respect to the other is the algebraic addition of all the component segments. **NOTE:** again, free-body diagram should be used to determine the internal resultant forces in different sections.

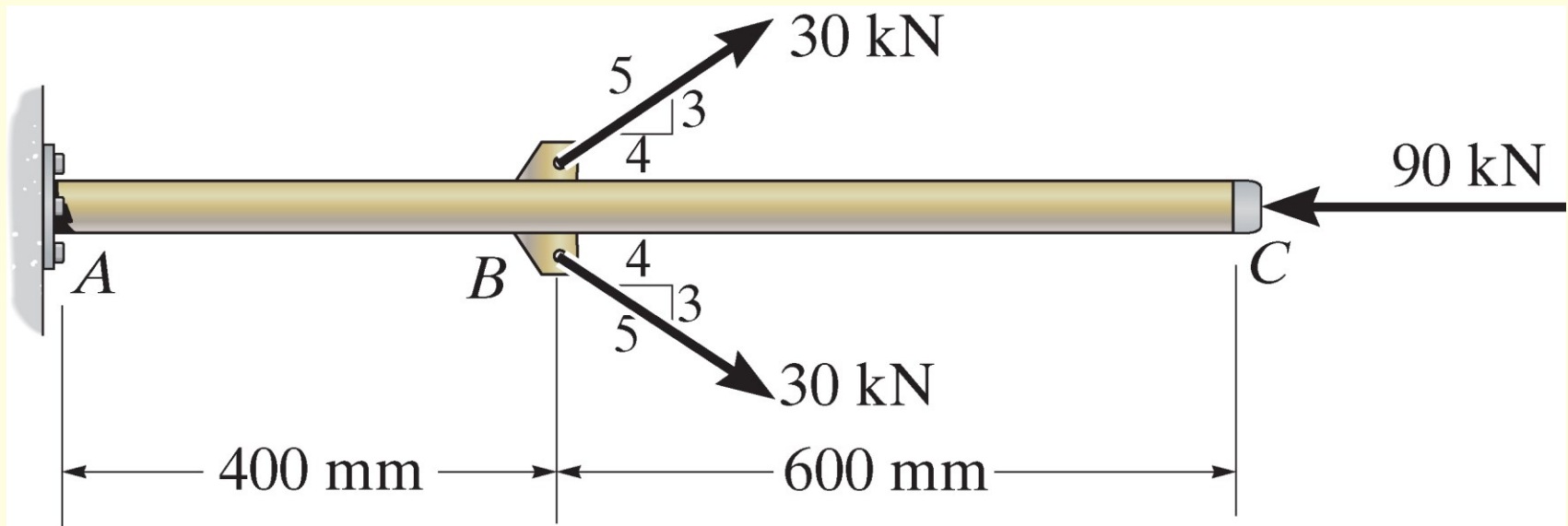


$$\delta = \int_0^L \frac{P(x)dx}{A(x)E(x)} = \frac{P_1 L_1}{A_1 E_1} + \frac{P_2 L_2}{A_2 E_2} + \dots = \sum \frac{P_i L_i}{A_i E_i}$$

Also refer to lecture ppt Ch. 3.1 page 8

IN-CLASS EXAMPLE 1

The 30-mm-diameter steel rod (modulus of elasticity is 200 GPa) is subject to the loading shown. Determine the displacement of point *B* and end *C*.



TIPS:

- (1) For axial loading, first determine the internal resultant force (stress) in each component segment by the section method
- (2) Second, use the Hooke's law to calculate the strain and/or displacement in each component
- (3) Third, the total displacement is the sum of displacement of all components

IN-CLASS EXAMPLE 1, cont'd

Solutions

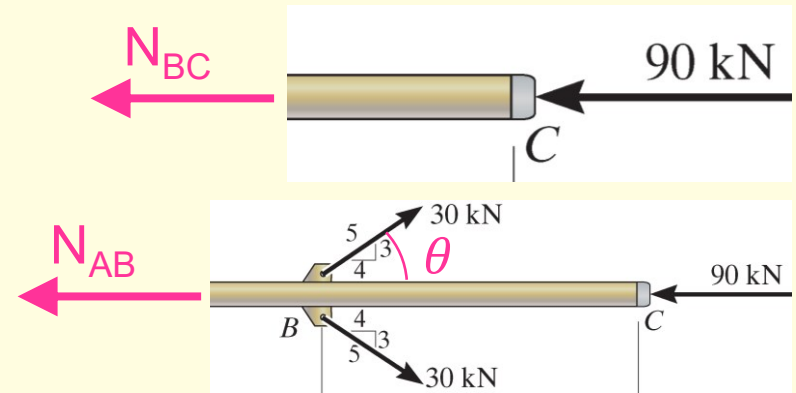
- Use the section method and free-body diagram to determine internal resultant forces

for segment BC

$$N_{BC} = -90 \text{ kN}$$

for segment AB

$$N_{AB} = -90 + 2 \times 30 \times \cos \theta = -42 \text{ kN}$$



NOTE:

We always assume the force and displacement is negative for compression and contraction. See textbook, page 127.

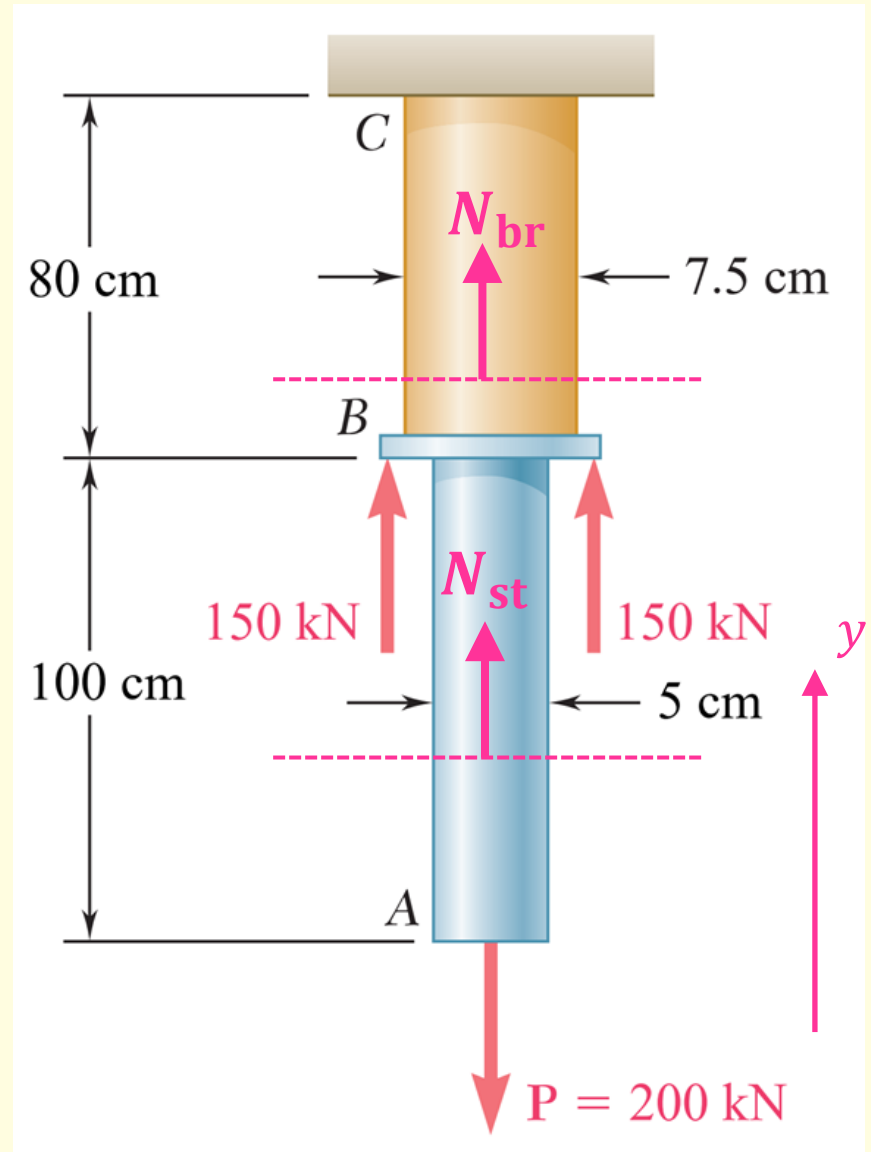
- Use the Hooke's law to calculate the strain and/or displacement

$$\delta_B = \frac{N_{AB} L_{AB}}{AE} = -\frac{42000 \times 0.4}{2 \times 10^{11} \times \frac{3.1415926}{4} \times 0.03^2} = -1.19 \times 10^{-4} \text{ m} = -0.119 \text{ mm} \quad (\text{Ans.})$$

$$\delta_C = \delta_B + \frac{N_{BC} L_{BC}}{AE} = -0.119 - \frac{90000 \times 0.6}{2 \times 10^{11} \times \frac{3.1415926}{4} \times 0.03^2} = -0.501 \text{ mm} \quad (\text{Ans.})$$

IN-CLASS EXAMPLE 2

Two cylindrical rods are joined at B and loaded as shown. Rod AB is made of steel (modulus of elasticity $E = 200 \text{ GPa}$), and rod BC of brass ($E = 105 \text{ GPa}$). Determine (a) the total deformation of the composite rod ABC , (b) the deflection of point B .



IN-CLASS EXAMPLE 2, cont'd

Solutions

Internal resultant force of steel and brass part:

Section the cylinder into two segments and apply free body diagram to the bottom segment:

$$\text{Steel} \qquad N_{st} - 200 = 0 \qquad \rightarrow \qquad N_{st} = 200 \text{ kN}$$

$$\text{Brass} \qquad N_{br} - 200 + 150 + 150 = 0 \qquad \rightarrow \qquad N_{br} = -100 \text{ kN}$$

Total deformation of the composite rod ABC :

$$\begin{aligned} \delta_{ABC} &= \delta_{AB} + \delta_{BC} = \frac{N_{st}L_{st}}{E_{st}A_{st}} + \frac{N_{br}L_{br}}{E_{br}A_{br}} \\ &= \frac{200000 \times 1}{200 \times 10^9 \times \frac{\pi}{4} \times 0.05^2} - \frac{100000 \times 0.8}{105 \times 10^9 \times \frac{\pi}{4} \times 0.075^2} \\ &= 3.37 \times 10^{-4} \text{ m} \\ &= 0.0337 \text{ cm} \qquad \textbf{(Ans.)} \end{aligned}$$

IN-CLASS EXAMPLE 2, cont'd

Solutions, cont'd

Since the point C is fixed, the deflection of point B should be equal to the displacement of segment BC :

$$\delta_B = \delta_{BC}$$

$$= \frac{N_{br} L_{br}}{E_{br} A_{br}}$$

$$= - \frac{100000 \times 0.8}{105 \times 10^9 \times \frac{\pi}{4} \times 0.075^2}$$

$$= -1.72 \times 10^{-4} \text{ m}$$

$$= -0.0172 \text{ cm} \quad \textbf{(Ans.)}$$

The negative value indicates the compressive strain in the segment BC.

RELEVANT PRACTICE

- Assignment 2 on Blackboard, problem #1 (go to Blackboard under the “Assignments” folder, **Assignment 2 due: 8:30 AM, Friday, Oct. 15th 2021**)

OUT-OF-CLASS READING

- **Textbook, pp. 123 – 132**
 - **In particular, Example 4.1 (pp. 129) and 4.3 (pp. 131)**